

GEN AI Chatbot — Documentation

BCG GenAI Job Simulation | Task 2

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Data Source: SEC EDGAR 10-K Filings — Microsoft (CIK 0000789019), Tesla (CIK 0001318605), Apple (CIK 0000320193)

1. How it Works

The chatbot utilizes keyword-based by using “if-else” to match the input to predefined financial queries. It allows users to convert three things:

- Which financial metrics or several at once or all metrics at once that they are querying (revenue, net income, and etc.)
- What company (Microsoft, Tesla, Apple)
- Which Year (2023, 2024, 2025)

Predefined queries can be indicated by following rules:

- What is the total revenue?
- What is Net Income?
- What is the sum of total assets?
- What is the sum of total liabilities?
- What is cash flow from operating activities?
- What is the revenue growth(%)?
- What is the net income growth(%)?
- What is the assets growth(%)?
- What is the liabilities growth(%)?
- What is the cash flow from operations growth(%)?
- What is the year by year average revenue growth rate(%)?
- What is the year by year average net income growth rate(%)?
- What is the year by year average assets growth rate(%)?
- What is the year by year average liabilities growth rate(%)?
- What is the year by year average cash flow from operations growth rate(%)?

2. What can be found in the Chatbot

- Microsoft: strongest financial performance across all metrics due to revenue growth from \$211,915M to \$281,724M with a year by year average revenue growth rate of 15.30%. In addition, Microsoft grew by 24.7% from 2023 to 2025 in AVG CFO.
- Apple: shows a stable and performance recovery as net income bounced back to \$112,010M in 2025 after a drop to \$93,736M in 2024. Apple also averaged 4.22% in revenue growth rate.
- Tesla: indicates margin compression as net income declined from \$14,974M in 2023 to \$3,855M in 2025 (-49.17% of income growth rate)

3. Limitations

- Predefined Queries only (entering number outside 1-15 leads to error)
- Keyword Matching-Based (chatbot only detects keywords inside the prompt, synonyms or other variations do not work)
- Processed Independently (Chat does not process previous messages)
- Limited Dataset (Limited to only 2023-2025 fiscal years, no live update)
- No machine learning or natural language included

4. How to run

- Open python ('GEN AI [TASK2.PY](#)')
- Type Hi to begin
- Select a year to query
- Enter a number from 1-15 or **mix the number** from 1-15 (1,7,10) or type '**all**'
- 'Reset' to Restart and 'Exit' to Quit

Author Information

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